

postscripts: This runs whenever a log is rotated in the configuration file specified block. The number of *postscript* executions for logs placed in the same directory can be overridden with *sharedscripts* directives.

Directives are also related to the intervals at which log files are rotated. They tell logrotate how often the log files should be rotated. The available options are:

1. Hourly (copy the file /etc/cron.daily/logrotate into the /etc/cron.hourly/ directory)
2. Daily
3. Weekly
4. Monthly
5. Yearly

Log files may also be rotated on the basis of file size. We can instruct logrotate to rotate files when the size of the file is greater than, let's say, 100KB, 100MB, 10GB, etc.

Some directives tell logrotate what number of rotated files to keep before deleting the old ones. In the following example, it will keep four rotated log files.

rotate 4

You can also use directives to remove rotated logs that are older than X number of days. The age is only checked if the log file is to be rotated. The files are mailed, instead of being deleted, to the configured address if *maillast* and *mail* are configured.

One can get the full list of commands used in logrotate configuration files by checking the man page:

man logrotate

Logrotate is one of the best utilities available in the Linux OS. It is ideal to take backups of applications, servers or any logs. By writing a script in the *postscript* section, we can move or copy backups of log files in Amazon S3 buckets as well. 

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Therefore, in the case of PaaS, cloudbursting is much easier than in IaaS. (Cloudbursting is the process of scaling out from private cloud to public cloud resources as per the load/demand on the application.)

8. DevOps orchestration and the API

DevOps can be defined in two ways:

1. It is a new name for automating the release management process that makes developers and the operations team work together.
2. The operations team manages operations by writing code, just like developers.

In DevOps, the application release management and application's resource demand management is of primary importance.

The traditional workflow tools like Jenkins have a new role of becoming orchestrators of all data centre components in an automated workflow. In this age of DevOps and ADDC, every product vendor releases the Jenkins plugins for their products as soon as it releases the product or its updates. This enables all of these ADDC components and the API endpoints to be orchestrated through a tool like Jenkins.

Apart from Jenkins, open source configuration management automation tools like Puppet and Chef can also easily integrate with other layers of ADDC to create a set of programmatic orchestration jobs exposed over API calls to run these jobs. These jobs can be run from API invocation, to orchestrate the data centre through the orchestration of all other API layers.

ADDC is therefore an approach to combining various independent technology solutions to create API endpoints for everything in a data centre. The benefit is the programmability

of the entire data centre. Theoretically, a program can be written to do all the jobs that are done by people in a traditional data centre. That is the automation nirvana which will be absolutely free of human errors and the most optimised process, because it will remove human elements from the data centre management completely. However, such a holistic app has not arrived yet. Various new age tools are coming up every day to take advantage of these APIs for specific use cases. So, once the data centre has been converted into an ADDC, it is only left to the developers' imagination as to how much can be automated – there is nothing that cannot be done.

Coming back to what we started with – the move towards architectures like ADDC is surely going to impact jobs as humans will be replaced by automation. However, there is the opportunity to become automation experts instead of sticking to manual labour profiles. Hence, in order to combat the new automation job role demands in the market, one needs to specialise in one or some of these ADDC building blocks to stay relevant in this transforming market. Hopefully, this article will help you build a mind map of all the domains you can try to skill up for. 

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